RAOH
Researchers

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Advancing knowledge.
Creating real-world impact.

Research Proposal

Definition

A research proposal is a structured scientific document that outlines a planned research study before it is conducted. It explains what the researcher intends to study, why the topic is important, and how the study will be carried out.

In simple terms, it is a **blueprint of the research project**.

Purpose of a Proposal

A research proposal is written to:

- Define a clear and focused research question
- Justify the importance of the study
- Describe the methodology in advance
- Obtain ethical approval (IRB)
- Get academic approval (e.g., university requirement)
- Plan resources, time, and feasibility

Importance at Yarmouk University (Medical Students)

At the Faculty of Medicine – Yarmouk University, the research proposal is typically required for:

- Graduation research projects
- Medical research methodology courses
- Ethical approval from the Institutional Review Board (IRB)
- Supervisor approval before starting data collection

It ensures that the study is:

- Scientifically valid
- Ethically acceptable
- Feasible within available time and resources

Structure of a Research Proposal

1. Title

Clear, specific, and reflects the research question.

2. Introduction / Background

- General information about the topic

- Current evidence from literature
- Identification of research gap
- Justification of the study

3. Problem Statement

A clear description of the exact problem the study addresses.

4. Objectives

- General objective
- Specific objectives

5. Hypothesis (if applicable)

A testable prediction about the relationship between variables.

6. Methodology

Includes:

- Study design
- Population and setting
- Sample size and sampling method
- Data collection tools
- Variables
- Statistical analysis
- Ethical considerations

7. Expected Results

Predicted outcomes based on literature (not actual results).

8. Timeline

Estimated duration for each phase of the study.

9. References

Properly formatted scientific citations.

Example (Medical Topic)

Title

Association Between Vitamin D Deficiency and Severity of Rheumatoid Arthritis Among Adult Patients in Northern Jordan

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Problem Statement

Rheumatoid arthritis is a chronic inflammatory disease associated with significant disability. Recent evidence suggests that vitamin D deficiency may be linked to increased disease severity; however, data from Jordanian populations remain limited.

Objective

To assess the association between serum vitamin D levels and disease severity in patients with rheumatoid arthritis.

Methodology (Summary Example)

- Study design: Cross-sectional study
- Population: Adult patients diagnosed with rheumatoid arthritis attending outpatient clinics in Northern Jordan
- Sample size: Calculated using OpenEpi
- Data collection: Clinical data and serum vitamin D levels extracted from medical records
- Analysis: SPSS, correlation and regression analysis

Steps to Write a Research Proposal

Step 1: Choose a Topic

Select a clinically relevant and feasible medical problem.

Step 2: Conduct Literature Review

Identify existing studies and research gaps.

Step 3: Formulate Research Question

Define a clear, focused question.

Step 4: Design Methodology

Choose study design and define population and variables.

Step 5: Write Proposal Draft

Organize all sections in a structured format.

Step 6: Review and Revise

Check clarity, feasibility, and scientific accuracy.

Step 7: Submit for Approval

Submit to supervisor and IRB if required.

Methods Used in Proposals

Common research designs used in proposals:

- Cross-sectional studies
- Case-control studies
- Cohort studies
- Randomized controlled trials
- Systematic reviews

Tips for Writing a Strong Proposal

- Choose a **feasible topic** (not too broad or complex)
- Ensure a **clear research gap**
- Use **recent and relevant references**
- Keep methodology realistic for student-level research
- Align objectives with methodology
- Avoid overly ambitious experimental designs
- Consult your supervisor early

Common Mistakes

- Vague research question
- Overly complex study design
- Missing methodology details
- No clear research gap
- Ignoring feasibility (time, sample, resources)
- Copying topics without originality

Golden Rule

A strong research proposal clearly answers:

What will you study? Why is it important? How will you study it?